

Research progresses of two kinds of novel electrolyte lithium salts:

lithium bis(oxalato) borate and lithium oxalyldifluoroborate

Tao Yu ^{1, a}, Hongming Zhang ^{2, b}, Xiaoli Xu ^{2, c}, Xiaoling Cui ^{2, d} and Liping Mao ^{2, e}

¹ Southwest Jiaotong University, Chengdu 610033, China

² Lanzhou University of Technology, Lanzhou 730050, China

^ayt001_yt@163.com, ^bzhang1989hming@163.com, ^cxiaoli562@126.com, ^dxlculw@163.com,
^ezyy871108@sina.cn

Keywords: electrolyte lithium salts; lithium oxalyldifluoroborate (LiODFB); lithium bis(oxalato) borate(LiBOB); electrochemical preformances

Abstract. Lithium bis(oxalato)borate (LiBOB) and lithium oxalyldifluoroborate (LiODFB) are promising boron-based lithium salts for lithium ion batteries. Advantages, disadvantages, and recent developments of each lithium salt were introduced respectively in this article. In addition, ionic conductivities and compatibilities with electrode materials were emphatically reviewed.

Introduction

Current lithium-ion batteries generally use lithium hexafluorophosphate (LiPF₆) in a blend of alkyl carbonates as the electrolyte. However, hazardous reactions of LiPF₆ were reported recently, because it is unstable and induces a cascade of unwanted side reactions and safety problems [1,2]. Recently, two lithium salts named lithium bis(oxalato)borate (LiBOB) and lithium oxalyldifluoroborate (LiODFB) have been reported as promising candidates for lithium-ion batteries with improved properties, such as good thermal stability, low cost and low susceptibility. Especially, it can effectively stabilize the graphite structure even in pure propylene carbonate (PC) and facilitate the formation of solid electrolyte interface (SEI) on the surface of carbonaceous anode materials [3–8]. Since LiODFB has the same oxalatoborate anhydride structure as LiBOB, most of the unique characteristics of LiBOB in lithium-ion batteries have been found from LiODFB [5], moreover, it can form a more stable SEI film with lower interface impedance on the surface of the anode materials.

Besides, ascribed to the unique chemical structure of LiODFB, the combined advantages of LiBOB and lithium tetrafluoroborate (LiBF₄) have been obtained from the LiODFB-based electrolyte, which include: (1) the optimized ionic conductivity over a wide temperature range, (2) the ability to provide a high cycling efficiency for metallic lithium plating and stripping on the surface of Cu, a current collector material of the graphite anode, (3) excellent ability to passivate Al at high potentials, (4) the ability to support graphite cycling in high PC-containing solutions, (5) the ability to support Li-ion cell operate at high temperatures, (6) the ability to support lithium ion batteries delivering high capacity at low temperatures and high current rates, and (7) the possibility to provide safety protection against abuse operations.

In this paper, the preparation methods and electrochemical performances of the two lithium salts LiBOB and LiODFB are discussed.

Using of the salts in lithium ion battery

The research progress of preparation of the two lithium salts

It has been reported that, LiBOB can be synthesized by LiOH, $\text{H}_2\text{C}_2\text{O}_4$ and H_3BO_3 in solvent water [9], however, it is an invalid method confirmed by later researchers, for the low purity of the product. Though high purity LiBOB can be synthesized by $\text{LiB}(\text{OCH}_3)_4$ and $(\text{CH}_3)_3\text{SiOCCOOSi}(\text{CH}_3)_3$ in anhydrous acetonitrile (CH_3CN) [10], the high cost of raw materials limits the generalization. In recent years, LiBOB is usually synthesized by LiOH, $\text{H}_2\text{C}_2\text{O}_4$ and H_3BO_3 with the ordinary solid phase method [11]. But the follow-up purification process must be carried out to meet the requirement of rechargeable lithium-ion battery, which greatly reduces the yield. LiBOB was synthesized in our laboratory by a compressing dry granulation method, which is cheaper, easier, more productive and more environmental benign than any previous method [12].

Relatively pure LiODFB could be synthesized by the chemical reaction between $\text{H}_2\text{C}_2\text{O}_4$ and LiBF_4 in AN solvent with the auxiliary reaction additives of SiCl_4 . The weight percentage content of unreacted LiBF_4 was decreased to 0.5%, but the cost was high [13]. To overcome these shortcomings, LiODFB product could be prepared with $(\text{C}_2\text{H}_5)_2\text{O} \cdot \text{BF}_3$ and $\text{Li}_2\text{C}_2\text{O}_4$ in dimethyl carbonate (DMC) solvent [14]. But half the lithium source of $\text{Li}_2\text{C}_2\text{O}_4$ was wasted on the by-product of LiF. And worse than that, the by-product of LiF would react with $(\text{C}_2\text{H}_5)_2\text{O} \cdot \text{BF}_3$ to form LiBF_4 , which would reduce the reaction selectivity. Besides, the unreacted LiF would coat upon the surface of the unreacted $\text{Li}_2\text{C}_2\text{O}_4$, which would prevent the reaction between $\text{Li}_2\text{C}_2\text{O}_4$ and $(\text{C}_2\text{H}_5)_2\text{O} \cdot \text{BF}_3$. LiODFB and LiBF_4 are simultaneously synthesized in our laboratory by an improved method with the purity of 99.5% and 99.5%, respectively, with a simple and cheap method, and find more appropriate solvent for separation and purification of LiODFB [15].

Electrochemical window of the salts and their thermal stability

LiBOB has a wide electrochemical window [>4.5 V] (vs. Li^+/Li), and can tolerate the high temperature (the thermal decomposition temperature of it is 302°C). The electrochemical window of LiODFB is lower than that of LiBOB [≥ 4.2 V] (vs. Li^+/Li), the thermal decomposition temperature of it begins at 240°C . The relatively lower decomposition temperature supplies a safe-keeping for misapplication (such as short circuit and discharge excessively). During overcharge, LiBOB produced only a small amount of smoke as long as the temperature did not exceed 100°C and it did not catch fire or spark, whereas LiPF_6 -based electrolytes not only caught fire, but resulted in a violent explosion and the maximum temperature reaching up to 400°C [16]. Besides, the current density of LiBOB and alkyl carbonate solvent electrolyte systems start to increase dramatically when the potential is higher than 4.5V, which due to LiBOB is instable at high potentials and first decomposed in electrolyte systems [12]. Fortunately, the LiBOB or LiODFB-based electrolyte could improve the stability of electrolyte system against oxidative decomposition with addition of sulfolane (SL), which the potential is higher than commercial electrolyte (5.5V vs. or 5.0V vs. Li^+/Li) [17].

Conductivities of the lithium salts

As LiBOB is of ideal solubility, electrolytes component with LiBOB and alkyl carbonate have high conductivity. For example, at 25°C , conductivity of 1.0 mol/L LiBOB/DME is 14.9×10^{-3} S/cm, conductivity of 1.0 mol/L LiBOB/AN is up to 25.2×10^{-3} S/cm and maximum solubility of LiBOB in pure GBL is about 2.5 mol/L. Besides, LiBOB is not sensitive to water, even though there sprinkle water in the electrolyte, no lithium salt hydrolytic decomposed is existed. Then, S. S. Zhang studied the

conductivity of 0.8 mol/L LiBOB, 1.0 mol/L LiBF₄ and 1.0 mol/L LiODFB/ PC+EC+EMC (1:1:3) [5]. When the temperature is higher than 10 °C, the order of conductivity is LiBOB> LiODFB>LiBF₄, but when it is lower than -30 °C, the order of conductivity becomes LiBF₄≈LiODFB> LiBOB. In other alkyl carbonate solvents, the three salts have the similar regularity. At the aspect of conductivity LiODFB combined the excellences of LiBOB and LiBF₄, and can maintain a superior conductivity in a more widely range of temperature, which in favour of broaden the using area of lithium-ion battery.

Compatibilities of LiBOB-based and LiODFB-based electrolytes with anode electrode materials

The solubility of LiBOB in the solvent of line alkyl carbonate is small, and the impedance of the SEI film formed on the surface of graphite is large, which lead to poor capacity consistent rate and high ratio discharge ability. Similar with LiBOB, LiODFB itself also could participate in formation of SEI film, even if in the pure PC, stable SEI film can also be formed on the surface of graphite. When mainly arising from large amounts of reductions of BOB or ODFB anions and organic solvents, which will typically lead to a thick SEI film with high impedance. Fortunately, the problem of high impedance of SEI film has been solved to some extent, due to the introduce of sulfone or sulfite in electrolyte systems, which to produce many compounds containing S element by electrochemical reduction and then precipitate on SEI layer [18-20]. In addition, the computational chemistry of molecular simulations has been used to gain understanding into the chemical composition of the electrode/electrolyte interface as a function of electrode potential. DFT calculations provide additional support for the structure of the LiBOB and LiDFOB derived components on the electrode surface [21,22]. Use of LiBOB as the electrolyte salt might prevent the dendrite formation on the Li metal anode when propylene carbonate is used as a plasticizer.

Compatibilities of LiBOB-based and LiODFB-based electrolytes with cathode electrode materials

LiBOB not only can passivation aluminum foil, but also shows excellent compatibility with LiFePO₄, which can reacts with Fe (II), the insoluble matters which are generated cover the surface of LiFePO₄, and then hinder the further reaction [12,23]. LiMn₂O₄, although with low cost, environment friendly, it restricte the dissolution of the manganese in LiPF₆-based electrolytes, which lead it difficult for commercial application. While, LiBOB is expected to solve the problem of dissolve erosion of Mn series cathode materials in electrolytes. Moreover, LiBOB can modify the high and low temperature performance of the battery. At 60 °C, LiMn₂O₄/Li battery with 0.8 mol/L LiBOB/EC+ EMC+DEC as the electrolytes is of good circularity and rate performance, especially the high rate performance. The capacity retention and capacity utilization at room temperature or 60 °C is higher than that of 1.0 mol/L LiPF₆/EC+ EMC+DEC electrolytes [25,26]. The cell performance such as discharge capacity and capacity retention of LiPF₆/LiBOB or LiBF₄/LiBOB mixtures, after long cycling at room temperature is much higher than that of each single salt. It is apparent that the salt blend is a good way to achieve superior cell performance [4,27-29].

Merged the advantages of LiBOB and LiBF₄, LiODFB reflects superior matching property with LiCoO₂, LiMn₂O₄, LiFePO₄, LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂ [30-33]. Compared with other lithium salts, battery with LiODFB-based electrolyte has superior high and low temperature performances and rate discharge ability. At 60 °C, using LiODFB as a electrolyte additive in LiFePO₄/graphite cell after 200 cycle the capacity retention is 66.4%, while, no LiODFB battery after 200 cycle the capacity retention is only 15%, the experiment proved that LiODFB as electrolyte additive can effectively improve LiFePO₄/graphite cell in high temperature conditions [21].

Conclusions

As a lithium salt, LiBOB has some unique advantages. It is hoped to replace LiPF_6 commerce partly or totally. Meanwhile, LiODFB combined the advantages of LiBF_4 and LiBOB, matched very well with anode and cathode electrode materials. Battery with it shows good electrochemical properties. But ,there are problems need to be solved before using in commerce, such as modify the synthesis technology, improve the purity of the salt, optimize the solvent system, and so on.

Acknowledgements

This work was supported by the Science and Technology Planning Project of Gansu Province (No. 110RJYA056) and the Scientific Research & Development Fund of Lanzhou University of Technology (No. SB05200903).

References

- [1] A. Hammami, N. Raymond and M. Armand: Nature (London) Vol. 424 (2003), p. 635
- [2] H. Markusson and P. Johansson: Electrochem. Solid-State Lett. Vol. 8 (2005), p. A215
- [3] G.V. Zhang, K. Xu and T.R. Jow: Electrochem. Solid-State Lett. Vol. 7 (2004), p. A224
- [4] K. Xu, S.S. Zhang and T.R. Jow: J. Power Sources Vol. 143 (2005), p. 197
- [5] S.S. Zhang: Electrochem. Commun Vol. 8 (2006), p. 1423
- [6] K. Xu, S.S. Zhang and B.A. Poesse: Electrochem. Solid-State Lett. Vol. 11 (2002), p. A259
- [7] K. Xu, S.S. Zhang and T.R. Jow: Electrochem. Solid-State Lett. Vol. 6 (2003), p. A117
- [8] K. Xu, U. Lee and S.S. Zhang: Electrochem. Solid-State Lett. Vol. 6 (2003), p. A144
- [9] U. Wietelmann, U. Lischka and M. Wegner, U. S. Patent US 6,506,516. (2003)
- [10] W. Xu, C. A. Angell, Electrochem. Solid-State Lett. Vol. 4 (2001), p. E1
- [11] B.T. Yu, W.H. Qiu, F.S. Li, Electrochem. Solid-State Lett. Vol. 9 (2006), p. A1
- [12] S.Y. Li, Y.Y. Zhao and X.M. Shi: Electrochim. Acta Vol. 65 (2012), p. 221
- [13] Herzig, C. Schreiner, D. Gerhard, P. Wasserscheid, H. J. Gores, J. Fluorine Chem. Vol. 128 (2007), p. 612
- [14] Zhang S S. J. Electrochem Commun. Vol. 8 (2006), p. 1423
- [15] S.Y. Li, W. Zhao and X.L. Cui: submitted to Electrochim. Acta
- [16] V. Aravindan, J. Gnanaraj and S. Madhavi: Chem. Eur Vol. 17 (2011), p. 14326
- [17] S.Y. Li, B.C. Li and X.L. Xu: J. Power Sources Vol. 209 (2012), p. 295
- [18] S.Y. Li, Y.Y. Zhao and X.M. Shi: Electrochim. Acta Vol. 65 (2012), p. 221
- [19] L.P. Mao, B.C. Li and X.L. Cui: Electrochim. Acta, Vol. 79 (2012), p. 197
- [20] S.Y. Li, X.L. Xu and X.M. Shi: J. Power Sources Vol. 217 (2012), p. 503
- [21] M.Q. Xu, L. Zhou and L.S. Hao: Journal of Power Sources Vol. 196 (2011), p. 6794
- [22] T. Yu, H.M. Zhang, S.Y. Li: submitted to J. Electrochem. Soc.

- [23] S.Y. Li, P.H. Ma and S.T. Song: Russ J Electrochem Vol. 44 (2008), p. 1144
- [24] B.T. Yu, W.H. Qiu and F.S. Li: J Power Sources Vol. 166 (2007), p. 499
- [25] Y.X. Guo, Z.G. Yin and Z.Y. Tao: J Power Sources Vol. 184 (2008), p. 513
- [26] M.H. Fu, K.L. Huang and S.Q. Liu: J Power Sources Vol. 195 (2010), p. 862
- [27] W. Xu, Z. Deng and X. Zhou, 210th ECS Meet. Abstr. Electrochem. Soc. Vol. 602 (2006), p. 251
- [28] E. Zinigrad, L.L. Asraf and J.S. Gnanaraj: 210th ECS Meet. Abstr. Electrochem. Soc. Vol. 602 (2006), p. 245
- [29] K. Xu, U. Lee and S. Zhang: 208th ECS Meet. Abstr. Electrochem. Soc. Vol. 502 (2006), p. 219
- [30] J. Li, K.Y. Xie and Y.Q. Xie: J Power Sources Vol. 195 (2010), p. 5344
- [31] Z.A. Zhang, X.J. Chen and F.Q. Li: J Power Sources Vol. 195 (2010), p. 7397
- [32] M.Q. Xu, L. Zhou and L.S. Hao: J Power Sources Vol. 196 (2011), p. 6794
- [33] H.Q. Gao, Z.A. Zhang and Y.Q. Lai: J Cent South Univ T Vol. 15 (2008), p. 830

Research Progresses of Two Kinds of Novel Electrolyte Lithium Salts: Lithium Bis(oxalato) Borate and Lithium Oxalyldifluoroborate

10.4028/www.scientific.net/AMR.652-654.896

DOI References

[1] A. Hammami, N. Raymond and M. Armand: Nature (London) Vol. 424 (2003), p.635.

<http://dx.doi.org/10.1038/424635b>

[4] K. Xu, S.S. Zhang and T.R. Jow: J. Power Sources Vol. 143 (2005), p.197.

<http://dx.doi.org/10.1016/j.jpowsour.2004.11.026>

[5] S.S. Zhang: Electrochem. Commun Vol. 8 (2006), p.1423.

<http://dx.doi.org/10.1016/j.elecom.2006.06.016>

[17] S.Y. Li, B.C. Li and X.L. Xu: J. Power Sources Vol. 209 (2012), p.295.

<http://dx.doi.org/10.1016/j.jpowsour.2012.03.004>

[18] S.Y. Li, Y.Y. Zhao and X.M. Shi: Electrochim. Acta Vol. 65 (2012), p.221.

<http://dx.doi.org/10.1016/j.electacta.2012.01.052>

[19] L.P. Mao, B.C. Li and X.L. Cui: Electrochim. Acta, Vol. 79 (2012), p.197.

<http://dx.doi.org/10.1016/j.electacta.2012.06.102>

[20] S.Y. Li, X.L. Xu and X.M. Shi: J. Power Sources Vol. 217 (2012), p.503.

<http://dx.doi.org/10.1016/j.jpowsour.2012.05.114>

[21] M.Q. Xu, L. Zhou and L.S. Hao: Journal of Power Sources Vol. 196 (2011), p.6794.

<http://dx.doi.org/10.1016/j.jpowsour.2010.10.050>

[23] S.Y. Li, P.H. Ma and S.T. Song: Russ J Electrochem Vol. 44 (2008), p.1144.

<http://dx.doi.org/10.1134/S1023193508100091>

[24] B.T. Yu, W.H. Qiu and F.S. Li: J Power Sources Vol. 166 (2007), p.499.

<http://dx.doi.org/10.1016/j.jpowsour.2007.01.038>

[25] Y.X. Guo, Z.G. Yin and Z.Y. Tao: J Power Sources Vol. 184 (2008), p.513.

<http://dx.doi.org/10.1016/j.jpowsour.2008.03.018>

[26] M.H. Fu, K.L. Huang and S.Q. Liu: J Power Sources Vol. 195 (2010), p.862.

<http://dx.doi.org/10.1016/j.jpowsour.2009.08.042>

[30] J. Li, K.Y. Xie and Y.Q. Xie: J Power Sources Vol. 195 (2010), p.5344.

<http://dx.doi.org/10.1016/j.jpowsour.2010.03.038>

[31] Z.A. Zhang, X.J. Chen and F.Q. Li: J Power Sources Vol. 195 (2010), p.7397.

<http://dx.doi.org/10.1016/j.jpowsour.2010.05.056>

[32] M.Q. Xu, L. Zhou and L.S. Hao: J Power Sources Vol. 196 (2011), p.6794.

<http://dx.doi.org/10.1016/j.jpowsour.2010.10.050>

[33] H.Q. Gao, Z.A. Zhang and Y.Q. Lai: J Cent South Univ T Vol. 15 (2008), p.830.

<http://dx.doi.org/10.1007/s11771-008-0153-1>